

OPTRONIC SENSORS

Night Vision Technologies



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Night vision devices are extensively used by the military for locating enemy targets, surveillance and navigation, thereby playing a crucial role in enhancing their night fighting capability. These are also used by law enforcement and security agencies for surveillance. Night vision-enabled cameras are also being used by both private business houses and military establishments to monitor surroundings of their critical assets.

Various approaches to night vision, their capabilities and limitations, a comparison between the two major categories of night vision devices, namely, image enhancement devices and thermal imaging devices, are presented in this part of the article.

Basic approaches to night vision

Night vision technologies and associated night vision devices enable users to see in low-light conditions. Contemporary devices allow viewing even in near total darkness.

The ability to see in low-light conditions is governed by two basic requirements: sufficient spectral range and sufficient intensity range. Low values of spectral range and intensity range are there in the case of human beings and, so, become the limiting factors for their ability to see with an acceptable level of contrast in low-light conditions. Use of technology to enhance both these parameters makes night vision possible.

Two basic and widely different approaches to night vision include image intensification (or enhancement) and thermal imaging. Both techniques are briefly described in the following paragraphs.

Image intensification. Image intensification or enhancement works on the principle of collecting small quanta of light reflected off the target scene, to be viewed in visible and near infrared (IR) bands of the electromagnetic spectrum in low-light conditions. Collected photons are amplified through the processes of photon-electron conversion, electron multiplication and electron-photon conversion. These processes take shape in an image intensifier tube.

Other important constituent parts of an image intensifier-tube-based night vision device are objective lens for collecting photons, eye piece for viewing the intensified image and

power supply for generating required DC voltages for electron acceleration.

Active illumination is often used in conjunction with an image intensifier tube in active night vision technology to enhance the image resolution in very low-light conditions. Illumination is generally provided by IR diodes emitting in the 700nm - 1000nm spectral band.

However, active night vision technology can be detected by night vision goggles and is therefore prone to giving away the location of the user. This is particularly undesirable in tactical military operations.

A variation of the conventional night vision device is the digital night vision device. While in the former, available light is collected through the objective lens and focused on an intensifier, the digital devices process and convert the optical image into an electric signal through a highly sensitive charge-coupled device (CCD) image sensor. This electrical signal is then transferred onto a micro-display, which is a type of LCD flat-panel display screen. The micro-display usually takes the form of an eyepiece, which can be used to view the image, instead of an LCD screen used in most digital cameras.

Digital night vision devices have relatively lower costs, are free from image distortions of photocathodes and blemishes of phosphorescent screen, are immune to damage by bright light exposure and offer image recording facility.

Thermal imaging. Thermal imaging night vision technology works on the principle of detecting the temperature differences between the objects in the foreground and those in the background. All objects above absolute zero temperature emit IR energy. Magnitude of the IR energy emitted by a hot body is proportional to fourth power of its absolute temperature (Stefan-Boltzmann law). Peak emission occurs at a wavelength that is inversely proportional to its absolute temperature (Wien's displacement law). Therefore hotter the body, higher the magnitude of IR energy emitted by the object and lower the wavelength of peak emission.

A thermal imaging device is essentially a heat sensor capable of detecting tiny differences in the temperature of different points on

Night vision technology has undergone considerable changes in the last 40 years. These changes have led to enormous improvements in performance standards of night vision devices